

About This Document

MCQs on machine learning, model evaluation and deep learning are reported across the assessment and interview.

Topics include overfitting, bias-variance, metrics, ensembles and neural-network basics.

Core Machine Learning Concepts

10 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Overfitting occurs when a model:

- (A) Performs poorly on both training and test data
- (B) Performs well on training data but poorly on unseen data ✓**
- (C) Has too few parameters
- (D) Cannot be trained

Answer: (B) Performs well on training data but poorly on unseen data

Q2. Which technique helps reduce overfitting?

- (A) Increasing model complexity
- (B) Regularization ✓**
- (C) Removing the validation set
- (D) Using fewer training epochs only

Answer: (B) Regularization

Q3. Supervised learning requires:

- (A) Unlabeled data
- (B) Labeled data ✓**
- (C) No data
- (D) Only test data

Answer: (B) Labeled data

Q4. Which of the following is an unsupervised learning algorithm?

- (A) Logistic Regression
- (B) K-means clustering ✓**
- (C) Decision Tree classifier
- (D) Linear Regression

Answer: (B) K-means clustering

Q5. In the bias-variance tradeoff, high bias typically leads to:

- (A) Overfitting
- (B) Underfitting ✓**
- (C) Perfect fit
- (D) Data leakage

Answer: (B) Underfitting

Q6. Which algorithm is an ensemble of many decision trees?

(A) K-means

(B) Random Forest ✓

(C) Naive Bayes

(D) KNN

Answer: (B) Random Forest

Q7. Principal Component Analysis (PCA) is used primarily for:

(A) Classification

(B) Dimensionality reduction ✓

(C) Clustering labels

(D) Data labeling

Answer: (B) Dimensionality reduction

Q8. Which is NOT a supervised learning algorithm?

(A) Linear Regression

(B) Logistic Regression

(C) K-means ✓

(D) Random Forest

Answer: (C) K-means

Q9. Boosting is an ensemble method that:

(A) Trains models in parallel independently

(B) Builds models sequentially, each correcting previous errors ✓

(C) Reduces the number of features

(D) Clusters the data

Answer: (B) Builds models sequentially, each correcting previous errors

Q10. Feature scaling (normalization/standardization) is most important for:

(A) Tree-based models

(B) Distance- and gradient-based algorithms ✓

(C) Rule-based systems

(D) Text tokenization

Answer: (B) Distance- and gradient-based algorithms

Model Evaluation

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Which metric is most appropriate for an imbalanced classification problem?

(A) Accuracy

(B) F1-score ✓

(C) Total count

(D) Learning rate

Answer: (B) F1-score

Q2. Precision is defined as:

(A) $TP / (TP + FN)$

(B) $TP / (TP + FP)$ ✓

(C) $TN / (TN + FP)$

(D) TP / Total

Answer: (B) $TP / (TP + FP)$

Q3. Recall (sensitivity) is defined as:

(A) $TP / (TP + FP)$

(B) $TP / (TP + FN)$ ✓

(C) $TN / (TN + FN)$

(D) $FP / (FP + TN)$

Answer: (B) $TP / (TP + FN)$

Q4. A confusion matrix shows:

(A) Feature correlations

(B) Counts of true/false positives and negatives ✓

(C) Model weights

(D) Learning curves

Answer: (B) Counts of true/false positives and negatives

Q5. Cross-validation is used to:

(A) Increase training data size

(B) Estimate how a model generalizes to unseen data ✓

(C) Reduce the number of classes

(D) Encode categorical features

Answer: (B) Estimate how a model generalizes to unseen data

Q6. The purpose of a train-test split is to:

(A) Speed up training

(B) Evaluate the model on data it has not seen during training ✓

(C) Balance the classes

(D) Remove outliers

Answer: (B) Evaluate the model on data it has not seen during training

Q7. Which loss function is typically used for regression?

(A) Binary cross-entropy

(B) Mean Squared Error ✓

(C) Hinge loss

(D) Categorical cross-entropy

Answer: (B) Mean Squared Error

Q8. Which loss function is typically used for binary classification?

(A) Mean Squared Error

(B) Binary cross-entropy ✓

(C) Mean Absolute Error

(D) R-squared

Answer: (B) Binary cross-entropy

Deep Learning and Preprocessing

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. The activation function most commonly used in the hidden layers of deep networks is:

(A) Sigmoid

(B) ReLU ✓

(C) Linear

(D) Step

Answer: (B) ReLU

Q2. The sigmoid activation function outputs values in the range:

(A) -1 to 1

(B) 0 to 1 ✓

(C) 0 to infinity

(D) -infinity to infinity

Answer: (B) 0 to 1

Q3. For multi-class classification, the output layer typically uses:

(A) ReLU

(B) Softmax ✓

(C) Tanh

(D) Linear

Answer: (B) Softmax

Q4. Gradient descent is an algorithm used to:

(A) Increase the loss

(B) Minimize the cost/loss function ✓

(C) Split the data

(D) Encode features

Answer: (B) Minimize the cost/loss function

Q5. If the learning rate is set too high, gradient descent may:

(A) Converge faster with no downside

(B) Fail to converge and overshoot ✓

(C) Always find the global minimum

(D) Stop updating weights

Answer: (B) Fail to converge and overshoot

Q6. One-hot encoding is used to handle:

(A) Missing numeric values

(B) Categorical variables ✓

(C) Outliers

(D) Imbalanced classes

Answer: (B) Categorical variables

Q7. A common technique for handling missing values in a dataset is:

(A) Encoding

(B) Imputation ✓

(C) Normalization

(D) Tokenization

Answer: (B) Imputation

Q8. L1 regularization (Lasso) tends to produce:

(A) Dense weight vectors

(B) Sparse models with some weights set to zero ✓

(C) Larger weights

(D) No change in weights

Answer: (B) Sparse models with some weights set to zero

Tredence

Machine Learning & Deep Learning

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

Version 1.0 • Last Updated: July 2026

Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

Table of Contents

About This Document	1
Core Machine Learning Concepts	1
Model Evaluation	2
Deep Learning and Preprocessing	4